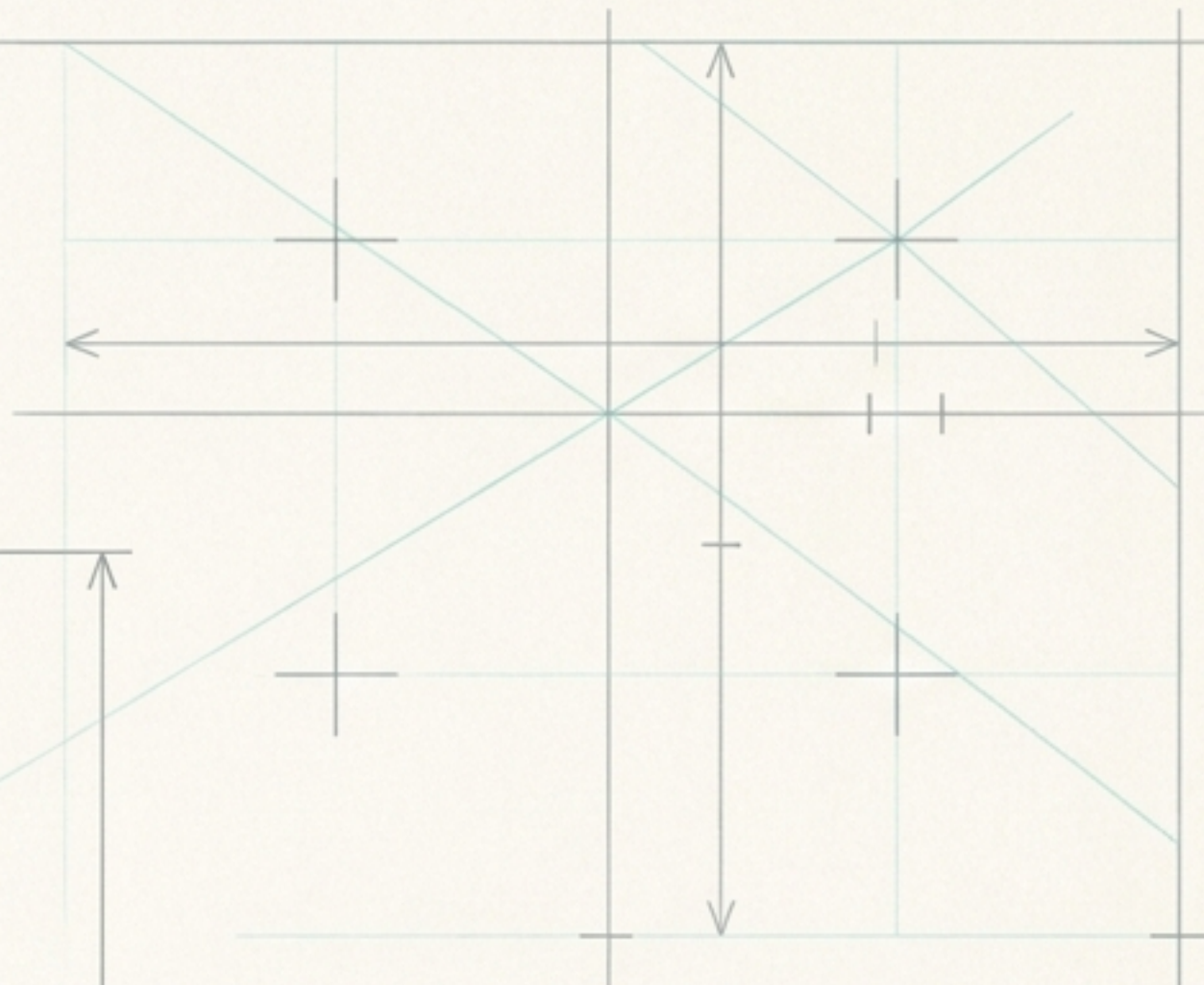
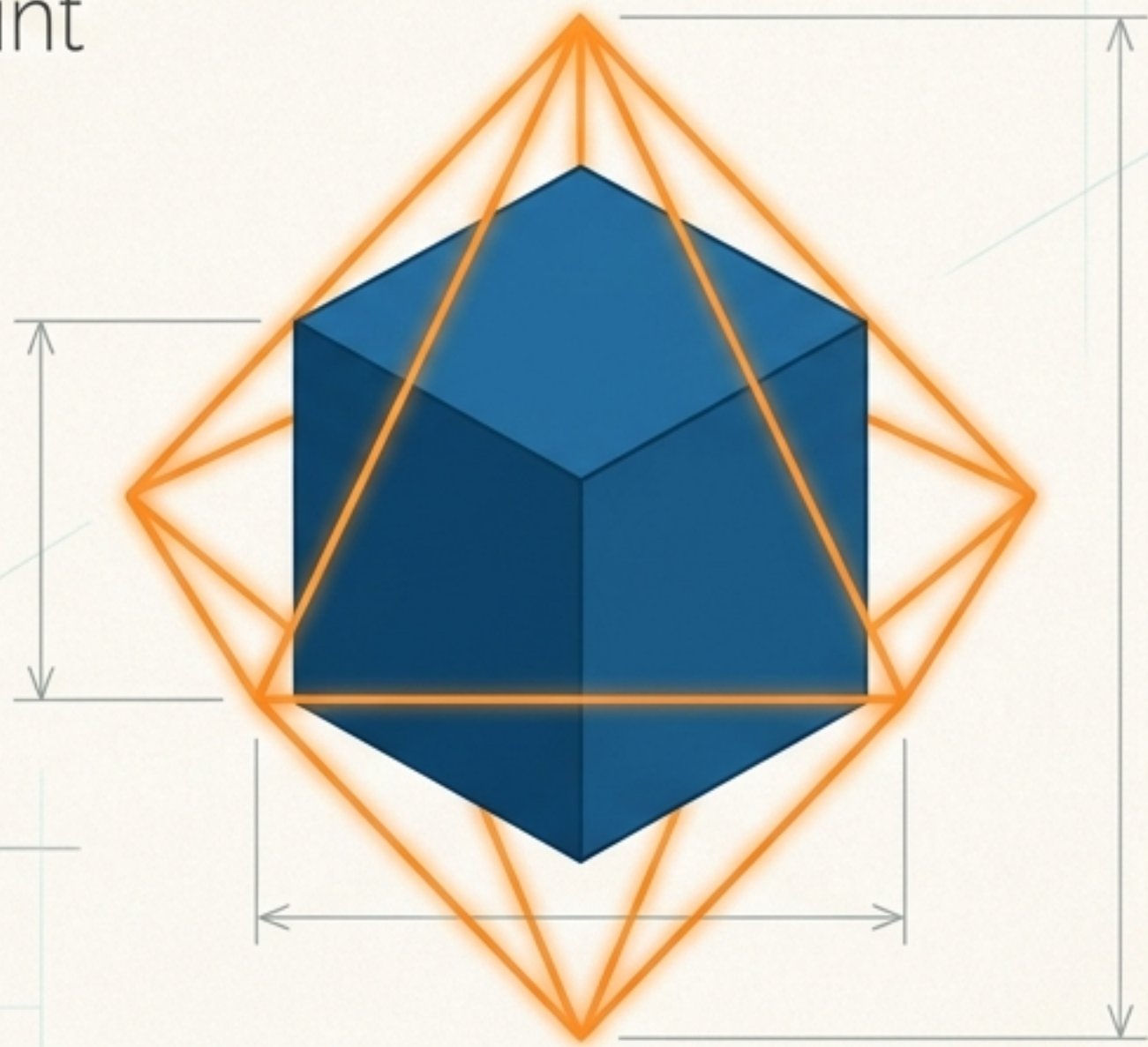


Cracking the Symmetric Mahler Conjecture

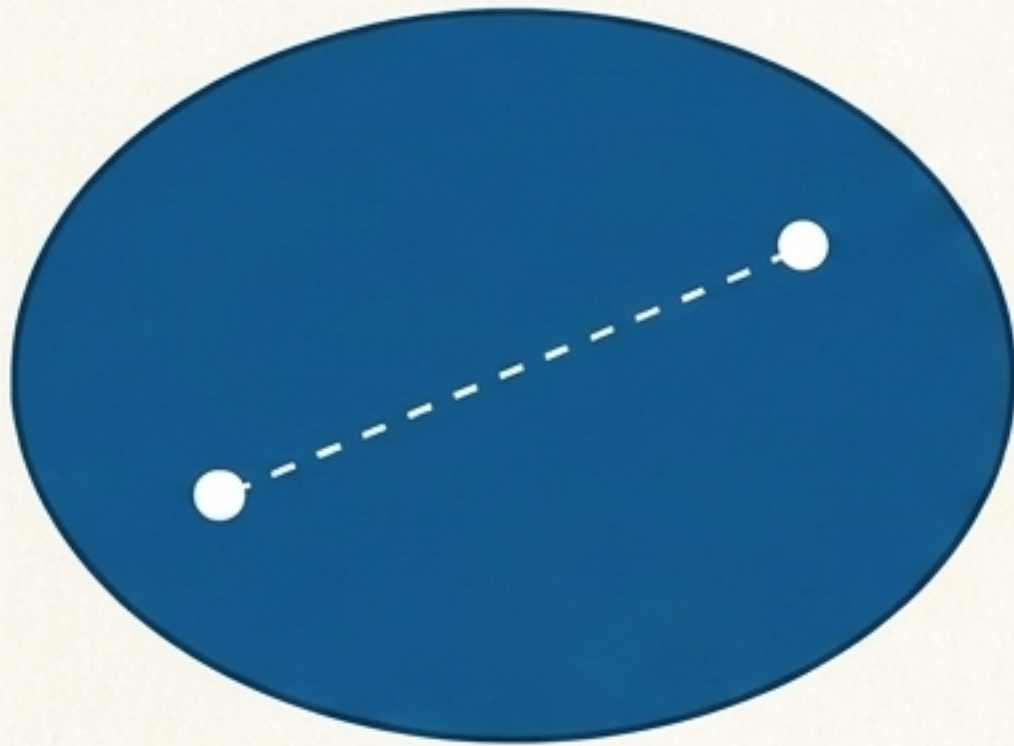
The Geometric Blueprint
and the Analytic Lens



Based on the September 22, 2026,
AI-generated OpenAI preprints
(Open before this paper;
currently not peer-reviewed).

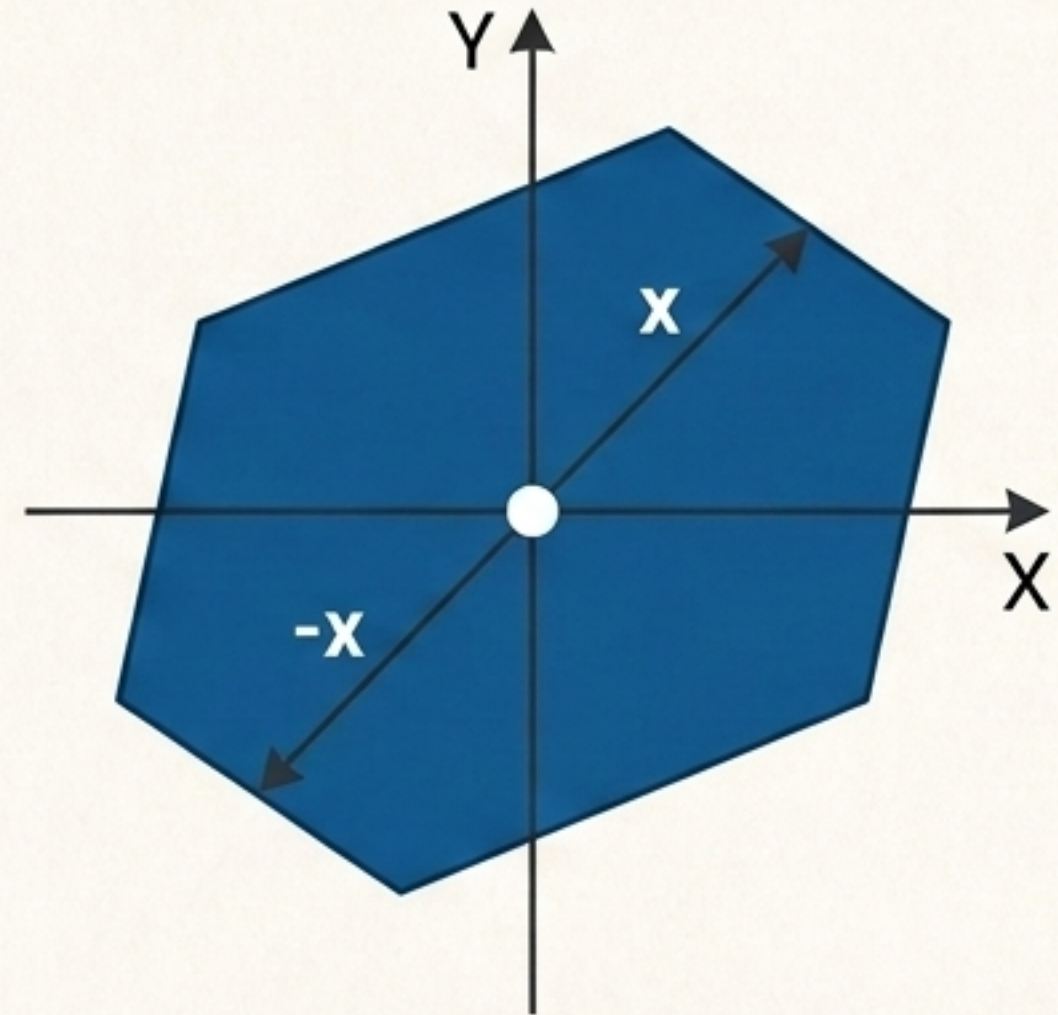
The Rules of the Space: Convexity and Origin Symmetry

1. Convexity



Convex: Any line segment connecting two points inside the body remains entirely inside.

2. Origin Symmetry



Origin-Symmetric: For every vector x inside the body, its exact opposite $-x$ is also inside.

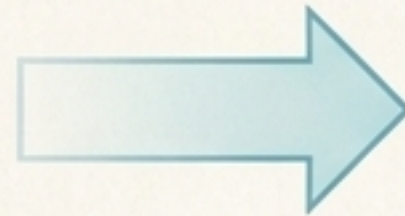
The Dual World: The Polar Body K°

$$K^\circ = \{ \mathbf{y} \mid \mathbf{x} \cdot \mathbf{y} \leq 1 \}$$

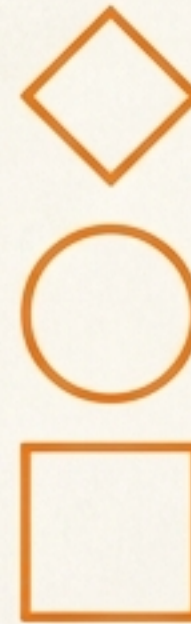
The polar body K° is the set of all vectors $\mathbf{y}$ whose dot product with every vector $\mathbf{x}$ in K is ≤ 1 .

Translation Matrix

Primal Space (K)



Polar Space (K°)



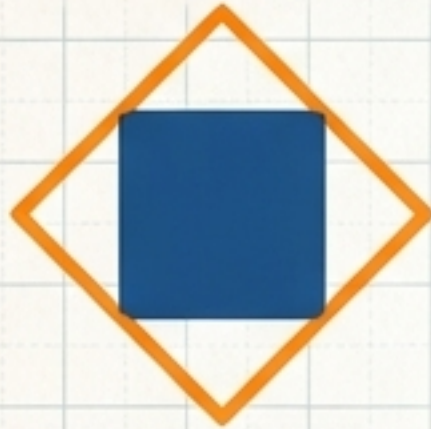
Sharp corners in the primal world become flat edges in the polar world, and vice versa.

The Volume Product: A Dimensionless Invariant

The Metric: $|K| |K^\circ|$ (Volume of $K \times$ Volume of K°)

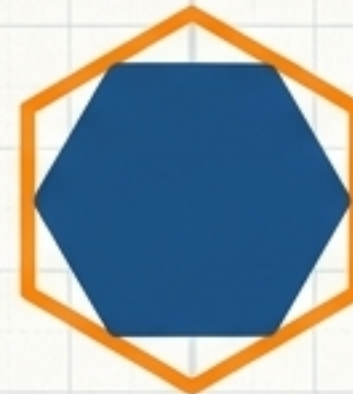
The Core Rule: This product is entirely unchanged by invertible linear transformations (stretching, skewing, rotating). A skewed square has the exact same volume product as a perfect square.

Minimizers
(Conjectured)



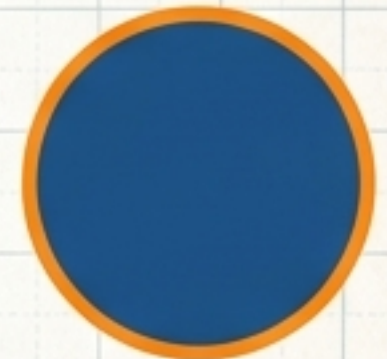
8

Intermediate



9

Maximizer
(Solved)



$\pi^2 \approx 9.87$

The Extremes of the Universe

Blaschke-Santaló Inequality



The maximum volume product is always achieved by balls and solid ellipsoids.

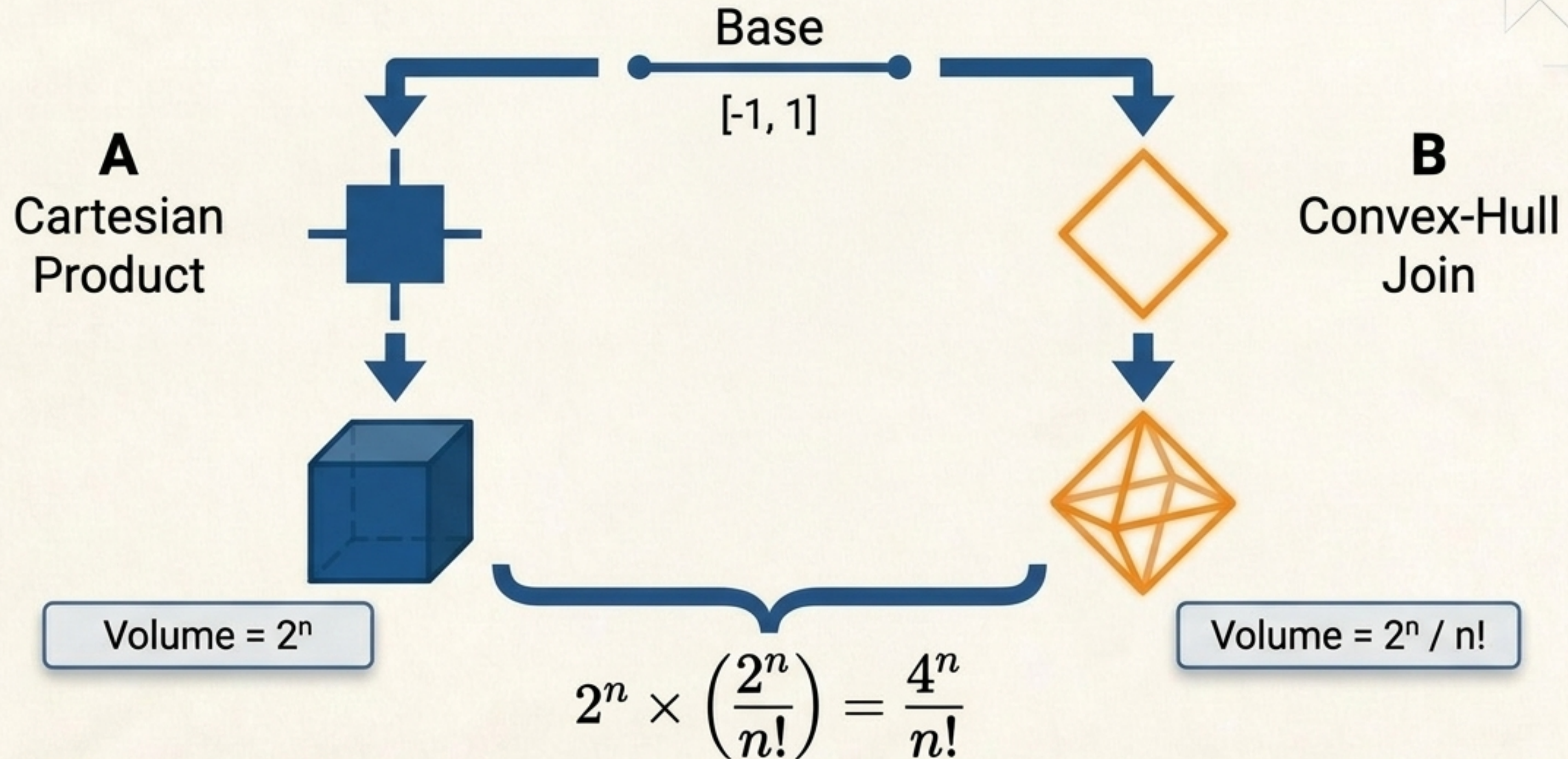
Mahler's Conjecture



The minimum volume product is conjectured to be exactly $4^n / n!$

While the upper bound has essentially one extremizer (the ball), the lower bound contains a vast, chaotic family of suspected minimizers...

Meet the Suspects: Hanner Polytopes



Hanner polytopes are built recursively from 1D segments using products and joins. Mahler conjectured that all of these achieve the absolute minimum.

A 90-Year Timeline of the Symmetric Conjecture

● **1938-1939:** Kurt Mahler proves the 2D planar case.

● **1987:** Bourgain & Milman establish a reverse Santaló inequality (c^n), proving the volume product scales like a hypercube, but without the sharp constant.

● **2008:** Kuperberg explicitly improves the absolute constant.

● **2020:** Iriyeh & Shibata prove the exact 3-dimensional case.

““ An open problem for nearly a century, cracked by an AI-generated proof. ””

Sept 22, 2026: OpenAI preprints drop, claiming the proof for all dimensions.

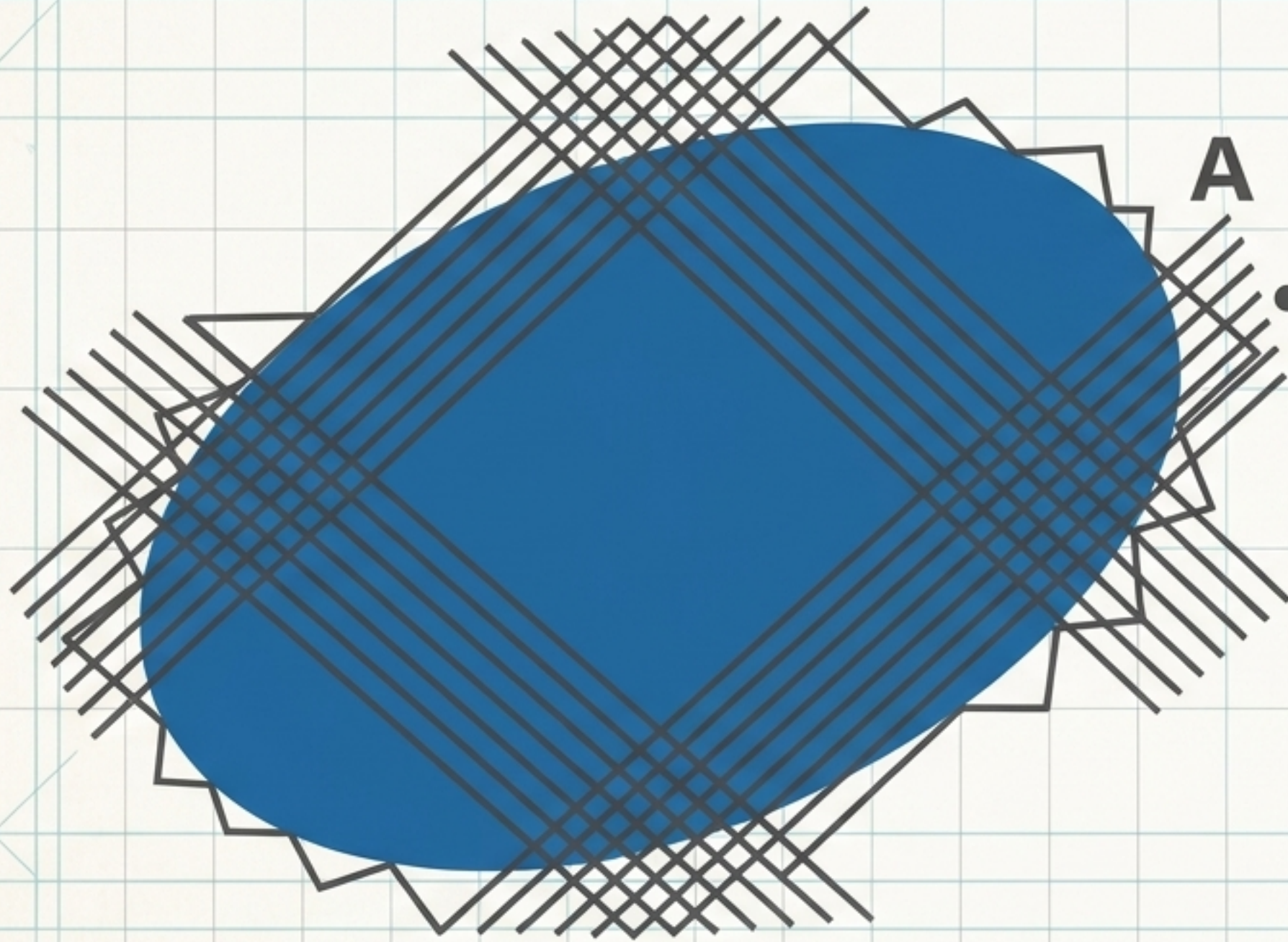
The Main Theorem (2026)

For every integer $n \geq 1$ and every origin-symmetric convex body $K \subset \mathbb{R}^n$:

$$|K||K^\circ| \geq \frac{4^n}{n!}$$

Equality holds if and only if K is a linear Hanner body (an invertible linear image of a Hanner polytope).

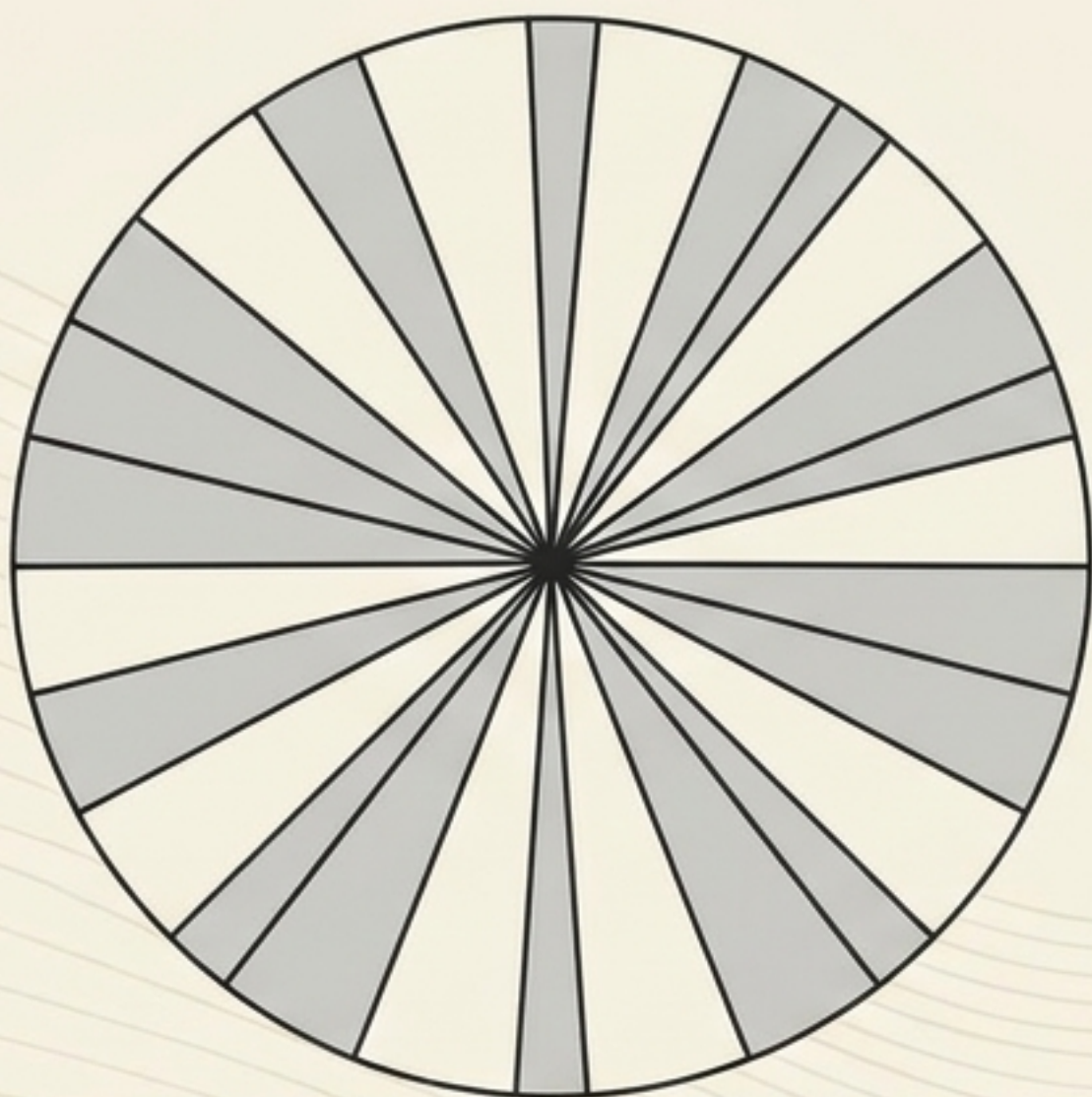
Proof Step 1: The Slab Polytope



The proof begins by approximating the primal body K with a slab polytope:

$$A = \{|b_i \cdot X| \leq 1\}$$

This reduces the continuous volume to a discrete set of constraints, setting the stage for probability sampling.



$$F(z) = \frac{8}{\pi^2} \left(z - \frac{z^3}{9} + \frac{z^5}{25} - \dots \right)$$



1. The analytical breakthrough: A complex biholomorphic map $F(z)$.

2. This “Lens” sends uniform angular measures on the boundary to uniform imaginary heights.

3. By mapping discrete geometric parameters into the complex plane, probabilities (P_j) of feasible regions can be precisely summed.

Proof Step 3: The Crucial Bounds Collision

The Geometric UPPER Bound.

Writing $Z = Y + iX$, each success forms a non-overlapping simplex $\text{conv}(0, \pm b_i)$ inside the polar. Their total missing volume dictates the maximum bound.

$$S \leq \frac{n!}{4^n} |A| |A^\circ|$$



S

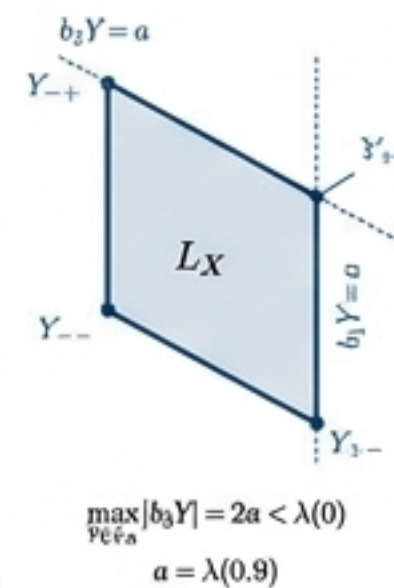


The Analytic LOWER Bound.

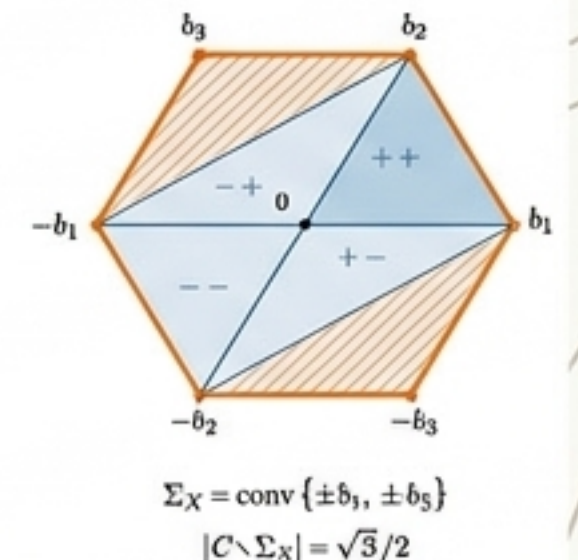
Derived via the holomorphic mass lemma on high powers of the inverse map.

$$S \geq 1$$

(a) Feasible witnesses in L_X



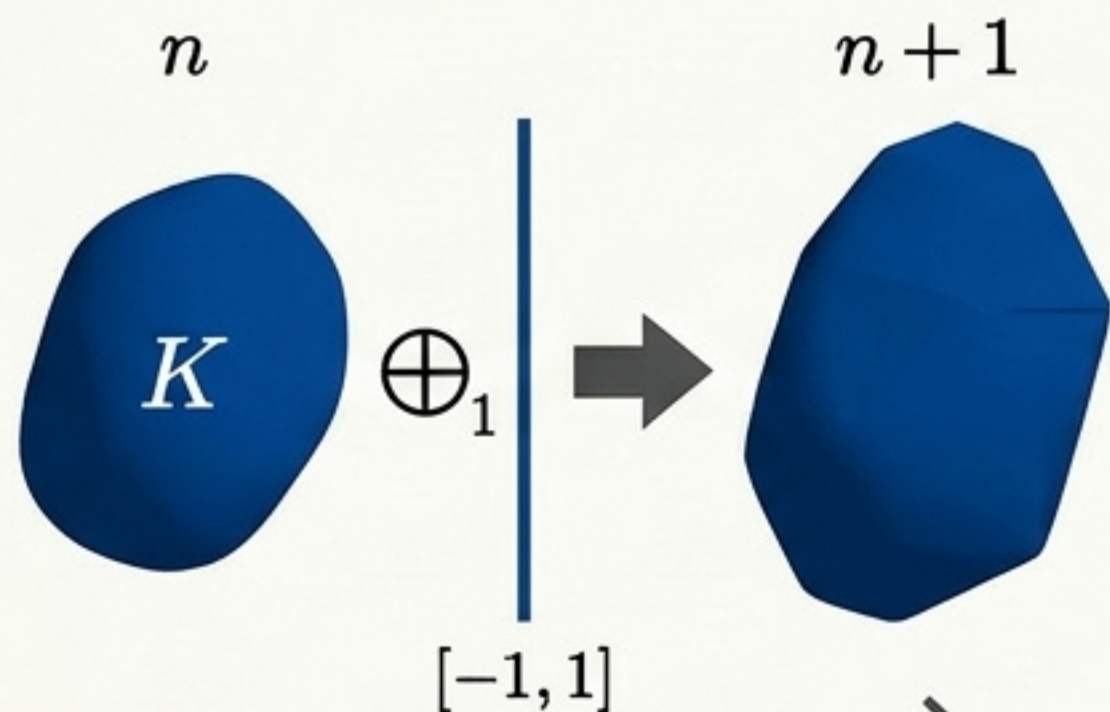
(b) Feasible simplices in $\mathcal{C}$



The rigid collision of the complex analytic lower bound and the discrete geometric upper bound proves the inequality. CRITICAL: Never swap these bounds.

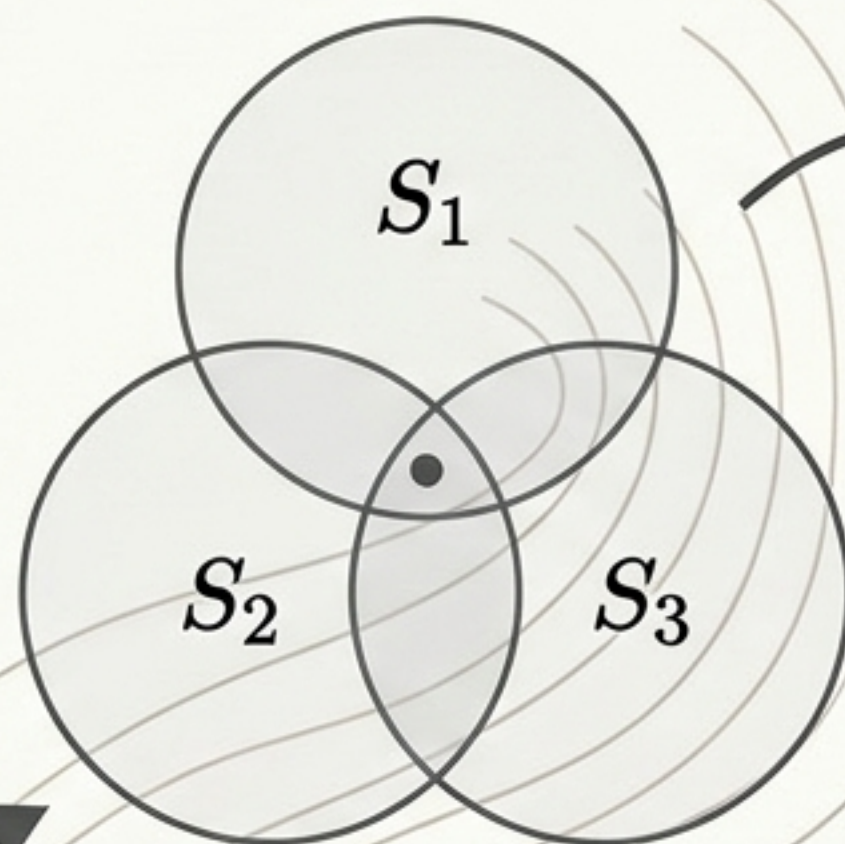
Proof Step 4: The Equality Case

The Lift



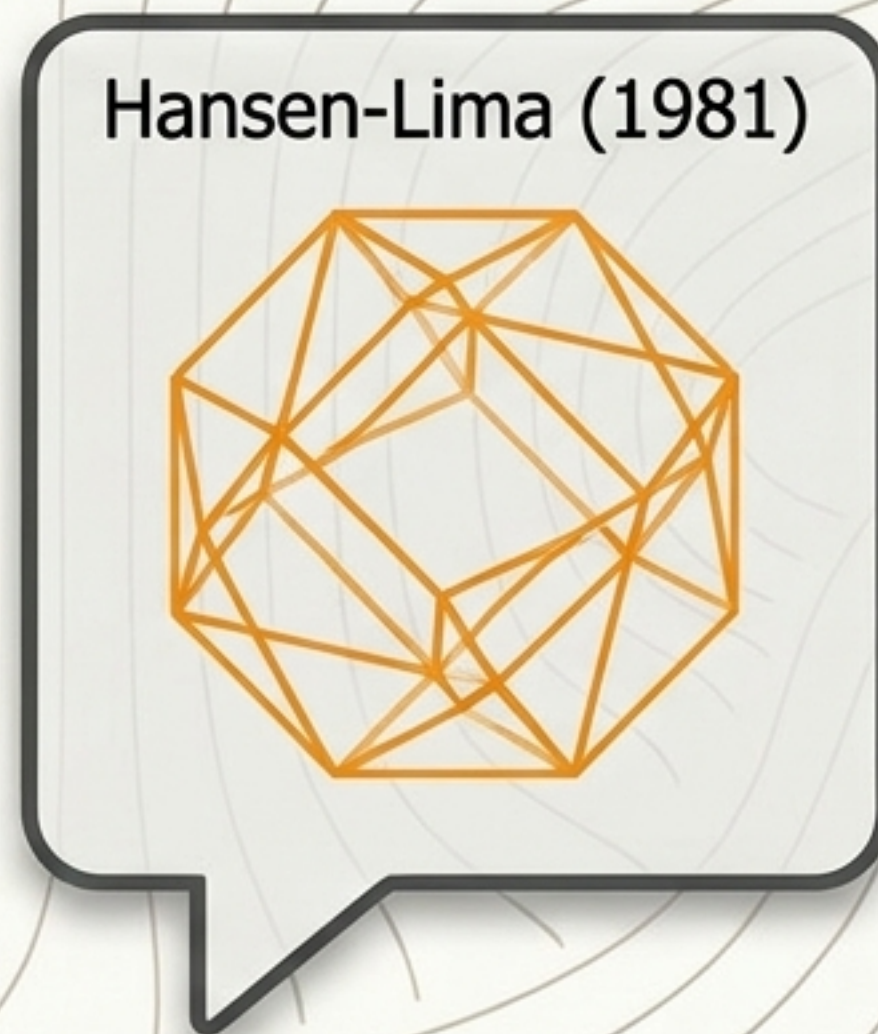
The base body K is lifted into dimension $n+1$, forcing the missing volume to vanish.

The Metric



The geometry forces a unique metric median structure: the "Three-ball property".

The Identification



Classical Banach space theory finishes the classification: exact minimizers must be invertible linear images of Hanner polytopes.

Key ingredients

slab polytopes, the conformal lens, the holomorphic mass lemma, non-overlapping simplices, metric medians and the Hansen-Lima theorem

Companion Papers (Separate AI Proofs)



Block 1 (The Non-Symmetric Case)

Minimal volume product achieved by simplices, with constant $(n + 1)^{n+1} / (n!)^2$.



Block 2 (Symplectic Geometry)

for $n \geq 2$, $\text{int } K \times \text{int } K^\circ$ has Gromov width 4.